CUSTOMER SEGMENTATION FOR E-COMMERCE

A Data Science Project using
EDA, SQL, K-Means Clustering & PCA

MAJOR PROJECT



SUBMITTED BY

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Introduction

Customer Segmentation is the process of dividing customers into groups based on similar purchasing behavior, spending habits, and demographic characteristics.

In the E-commerce industry, understanding customer behavior is essential for improving marketing strategies, increasing customer satisfaction, and maximizing business revenue.

This project uses Data Science techniques such as **Exploratory Data Analysis (EDA), SQL, Machine Learning (K-Means & DBSCAN), Principal Component Analysis (PCA), and Data Visualization** to identify meaningful customer segments and provide actionable business insights.

Problem Statement

E-commerce businesses collect large volumes of customer data, but analyzing this information effectively is a major challenge.

Challenges

- Understanding customer purchasing behavior
- Identifying valuable customers
- Improving customer retention
- Reducing ineffective marketing campaigns
- Delivering personalized recommendations

Proposed Solution

Develop a customer segmentation system using clustering algorithms to group customers with similar characteristics and support data-driven business decisions

Objectives

- Perform Data Cleaning & Preprocessing
- Conduct Exploratory Data Analysis (EDA)
- Analyze customer data using SQL
- Apply K-Means Clustering
- Apply DBSCAN Clustering
- Reduce dimensionality using PCA
- Visualize customer segments
- Generate business recommendations for targeted marketing

Project Methodology

Step 1
Data Collection
↓
Step 2
Data Cleaning & Preprocessing
↓
Step 3
SQL Analysis
↓
Step 4
Exploratory Data Analysis (EDA)
↓
Step 5
Feature Selection & Standardization
↓
Step 6
K-Means Clustering
↓
Step 7
DBSCAN Clustering
↓
Step 8
Principal Component Analysis (PCA)
↓
Step 9
Visualization
↓
Step 10
Business Insights

Technologies Used

Programming Language

- Python

Database

- SQLite (SQL)

Libraries

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

Machine Learning

- K-Means
- DBSCAN
- PCA

IDE

- Jupyter Notebook
- VS Code

Dataset Description

The project uses an E-commerce customer dataset containing customer demographic and spending information.

Dataset Features

- Customer ID
- Gender
- Age
- Annual Income (k$)
- Spending Score (1–100)

Data Preprocessing

- Missing Value Handling
- Duplicate Removal
- Data Cleaning
- Feature Scaling
- Data Standardization

Exploratory Data Analysis (EDA)

EDA was performed to understand customer characteristics before applying machine learning models.

Analysis Performed

- Gender Distribution
- Age Distribution
- Annual Income Distribution
- Spending Score Distribution
- Correlation Analysis
- Outlier Detection
- Missing Value Analysis

Visualizations

- Histograms
- Boxplots
- Heatmaps
- Pairplots
- Scatter Plots

SQL Analysis

SQL queries were used to analyze customer information and generate business insights.

Sample Queries

- Total Number of Customers
- Average Annual Income
- Average Spending Score
- Customer Distribution by Gender
- Highest Spending Score
- Income-wise Customer Analysis
- Customer Count by Age Group

Benefits

- Efficient Data Retrieval
- Faster Business Reporting
- Better Decision-Making

Customer Segmentation

K-Means Clustering

- Groups similar customers
- Uses Elbow Method for optimal clusters
- Fast and efficient

DBSCAN

- Density-based clustering
- Detects outliers automatically
- Handles irregular cluster shapes

PCA

Principal Component Analysis reduces high-dimensional data into fewer dimensions while preserving important information.

Benefits

- Reduces Complexity
- Faster Processing
- Better Visualization
- Improves Cluster Interpretation

Output

- 2D PCA Projection
- Cluster Scatter Plot

Project Structure

```
Customer-Segmentation-Ecommerce
│
├── data
├── notebooks
├── sql
├── models
├── visualizations
├── requirements.txt
└── README.md
```

Major Modules

- Data Loading
- Data Cleaning
- SQL Queries
- EDA
- K-Means Model
- DBSCAN Model
- PCA
- Visualization

Results

The clustering algorithms successfully grouped customers based on **Age, Annual Income, and Spending Score**.

Generated Outputs

- Customer Clusters
- PCA Projection
- Heatmap
- Histogram
- Boxplot
- Scatter Plot
- Cluster Visualization

Key Observations

- High-income customers do not always have high spending scores.
- Customers with similar income levels often exhibit different spending behaviors.
- K-Means created well-defined customer groups for marketing.
- DBSCAN effectively identified outliers in the dataset.
- PCA simplified visualization while preserving most of the important information.

Business Recommendations

- **High Spending Customers:** Offer premium memberships and exclusive rewards.
- **Moderate Spending Customers:** Provide loyalty programs and personalized offers.
- **Low Spending Customers:** Attract them through discounts and promotional campaigns.
- **Potential Customers:** Design targeted marketing strategies based on age and income segments.

Future Improvements

- Hyperparameter Tuning
- Power BI Dashboard
- Streamlit Web Application
- Real-Time Customer Segmentation
- Customer Lifetime Value Prediction
- Customer Churn Prediction
- Recommendation System
- Cloud Deployment

Conclusion

This project successfully demonstrates how **EDA, SQL, Machine Learning, and PCA** can transform raw customer data into meaningful customer segments.

The generated customer groups help businesses:

- Understand customer behavior
- Improve marketing strategies
- Increase customer retention
- Maximize sales and profitability

The project achieves all the defined objectives and provides a strong foundation for real-world E-commerce analytics.